

MailerLite Debugging Details

Error Message Analysis

User-Reported Error

Email sending failed: The given data was invalid. Results saved locally.

API Response Details

- **HTTP Status:** 422 Unprocessable Entity (inferred from error message)
- **Error Message:** "The given data was invalid"
- **Context:** Error occurred when creating/updating subscribers in MailerLite

Data Flow Analysis

Request Path

```
User completes quiz
→ Frontend (quiz-client.tsx) sends POST to /api/mailerlite
→ API Route (app/api/mailerlite/route.ts) validates data
→ Calls createOrUpdateSubscriber() in lib/mailerlite-service.ts
→ Makes POST request to https://connect.mailerlite.com/api/subscribers
→ MailerLite API validates request structure
→ ❌ REJECTED: Invalid data format
```

The Bug in Detail

Code Location: `lib/maillite-service.ts` (Lines 32-52)

✗ BEFORE (Incorrect)

```
const subscriberData = {
  email: submission.email,
  fields: {
    name: submission.name, // ✗ BUG: name is a default field, not custom!
    total_score: submission.totalScore.toString(),
    max_score: submission.maxScore.toString(),
    result_type: submission.resultType,
    recommendations: recommendationsText,
    submission_date: submission.timestamp,
    willingness_to_change_score: submission.categoryScores.veraenderungsbereitschaft?.toFixed(1) || 'N/A',
    adaptability_score: submission.categoryScores.anpassungsfahigkeit?.toFixed(1) || 'N/A',
    risk_tolerance_score: submission.categoryScores.risikobereitschaft?.toFixed(1) || 'N/A',
    financial_situation_score: submission.categoryScores.finanzielle_situation?.toFixed(1) || 'N/A',
    value_compass_score: submission.categoryScores.wertekompass?.toFixed(1) || 'N/A',
    need_for_security_score: submission.categoryScores.sicherheitsbeduerfnis?.toFixed(1) || 'N/A',
    growth_vs_comfort_score: submission.categoryScores.growth_vs_komfort?.toFixed(1) || 'N/A',
    conformity_vs_rebel_score: submission.categoryScores.konformitaet_vs_rebell?.toFixed(1) || 'N/A',
  },
  groups: [MAILERLITE_GROUP_ID],
};
```

Why This Failed:

- MailerLite's API rejected the request because `name` is a **default field**
- Default fields must be at the **root level** of the JSON payload
- Only **custom fields** (created via API or dashboard) go inside the `fields` object
- The API validation failed: "The given data was invalid"

✓ AFTER (Fixed)

```
const subscriberData = {
  email: submission.email,
  name: submission.name, // ✓ FIX: Moved to top level (default field)
  fields: {
    // Only custom fields here
    total_score: submission.totalScore.toString(),
    max_score: submission.maxScore.toString(),
    result_type: submission.resultType,
    recommendations: recommendationsText,
    submission_date: submission.timestamp,
    willingness_to_change_score: submission.categoryScores.veraenderungsbereitschaft?.toFixed(1) || 'N/A',
    adaptability_score: submission.categoryScores.anpassungsfahigkeit?.toFixed(1) || 'N/A',
    risk_tolerance_score: submission.categoryScores.risikobereitschaft?.toFixed(1) || 'N/A',
    financial_situation_score: submission.categoryScores.finanzielle_situation?.toFixed(1) || 'N/A',
    value_compass_score: submission.categoryScores.wertekompass?.toFixed(1) || 'N/A',
    need_for_security_score: submission.categoryScores.sicherheitsbeduerfnis?.toFixed(1) || 'N/A',
    growth_vs_comfort_score: submission.categoryScores.growth_vs_komfort?.toFixed(1) || 'N/A',
    conformity_vs_rebel_score: submission.categoryScores.konformitaet_vs_rebell?.toFixed(1) || 'N/A',
  },
  groups: [MAILERLITE_GROUP_ID],
};
```

Why This Works:

- name is now at the root level where MailerLite expects it
- All custom fields remain properly nested in the fields object
- The API validation passes: HTTP 201 (Created) response



MailerLite API Structure Reference

Correct JSON Structure for POST /subscribers

```
{
  "email": "user@example.com",           // Required, top level
  "name": "John Doe",                   // Optional, top level (default field)
  "status": "active",                   // Optional, top level (default field)
  "fields": {                           // Custom fields object
    "total_score": "50",                 // Custom field
    "max_score": "75",                   // Custom field
    "result_type": "Well Prepared",      // Custom field
    "recommendations": "• Item 1",       // Custom field
    "submission_date": "1/15/2026",     // Custom field
    // ... more custom fields
  },
  "groups": ["176599002499778006"]      // Group IDs array
}
```

Testing Results

Test 1: Original Test Script

```
$ node test-mailerlite.js
```

Before Fix:

- ❌ Status: 422 or 400 (Unprocessable Entity / Bad Request)
- ❌ Error: "The given data was invalid"

After Fix:

- ✅ Status: 201 (Created)
- ✅ Subscriber ID: 176624345421972896
- ✅ All fields populated correctly

Test 2: Standalone Verification

```
$ node test-fix-standalone.js
```

Result:

- ✅ Status: 201 (Created)
- ✅ Subscriber ID: 176624502649652630
- ✅ Test used same name as error screenshot ("QQQQQQQQQQ")

Test 3: Build Verification

```
$ npm run build
```

Result:

- ✅ Compiled successfully
- ✅ No TypeScript errors
- ✅ All routes valid

Request/Response Comparison

Before Fix - Failed Request

```
POST https://connect.mailerlite.com/api/subscribers HTTP/1.1
Content-Type: application/json
Authorization: Bearer [API_KEY]
```

```
{
  "email": "test@example.com",
  "fields": {
    "name": "QQQQQQQQQ",  WRONG LOCATION
    "total_score": "50",
    ...
  },
  "groups": ["176599002499778006"]
}
```

```
Response: 422 Unprocessable Entity
{
  "message": "The given data was invalid"
}
```

After Fix - Successful Request

```
POST https://connect.mailerlite.com/api/subscribers HTTP/1.1
Content-Type: application/json
Authorization: Bearer [API_KEY]
```

```
{
  "email": "test@example.com",
  "name": "QQQQQQQQQ",  CORRECT LOCATION
  "fields": {
    "total_score": "50",
    "max_score": "75",
    ...
  },
  "groups": ["176599002499778006"]
}
```

```
Response: 201 Created
{
  "data": {
    "id": "176624502649652630",
    "email": "test@example.com",
    "status": "active",
    "fields": {
      "name": null,  Note: name appears here as null (it's at top level)
      "total_score": "50",
      "max_score": "75",
      ...
    }
  }
}
```

Lessons Learned

1. **Default vs Custom Fields:** Always check MailerLite documentation for which fields are default

2. **API Structure:** Default fields go at root level, custom fields go in `fields` object
3. **Error Messages:** “Invalid data” errors often indicate structural issues, not value issues
4. **Testing:** Use standalone scripts to test API integration independent of app complexity
5. **Documentation:** The field creation script showed `name` was NOT in the custom fields list

How This Was Discovered

1. Reviewed the error message and context
2. Examined the field creation script (`scripts/create-mailerlite-fields.js`)
3. Noticed `name` was NOT in the list of custom fields to create
4. Realized `name` must be a default MailerLite field
5. Checked the test script and saw it also had `name` in wrong location
6. Fixed the structure to match MailerLite API requirements
7. Verified with successful API tests

Verification Checklist

- ☒ [x] Code fix applied to `lib/mailerlite-service.ts`
 - ☒ [x] Test script updated to match correct format
 - ☒ [x] Standalone test passes (HTTP 201)
 - ☒ [x] Original test script passes (HTTP 201)
 - ☒ [x] Build compiles successfully
 - ☒ [x] Git commit created with clear message
 - ☒ [x] Documentation created (this file + `MAILERLITE_FIX_SUMMARY.md`)
 - ☐ [] Real quiz submission tested (requires running app)
 - ☐ [] MailerLite dashboard verified (requires access)
-

Fix Date: January 15, 2026

Debug Duration: ~30 minutes

Root Cause: Field structure mismatch (default vs custom fields)

Solution: 1-line change (move `name` field to correct location) + comments